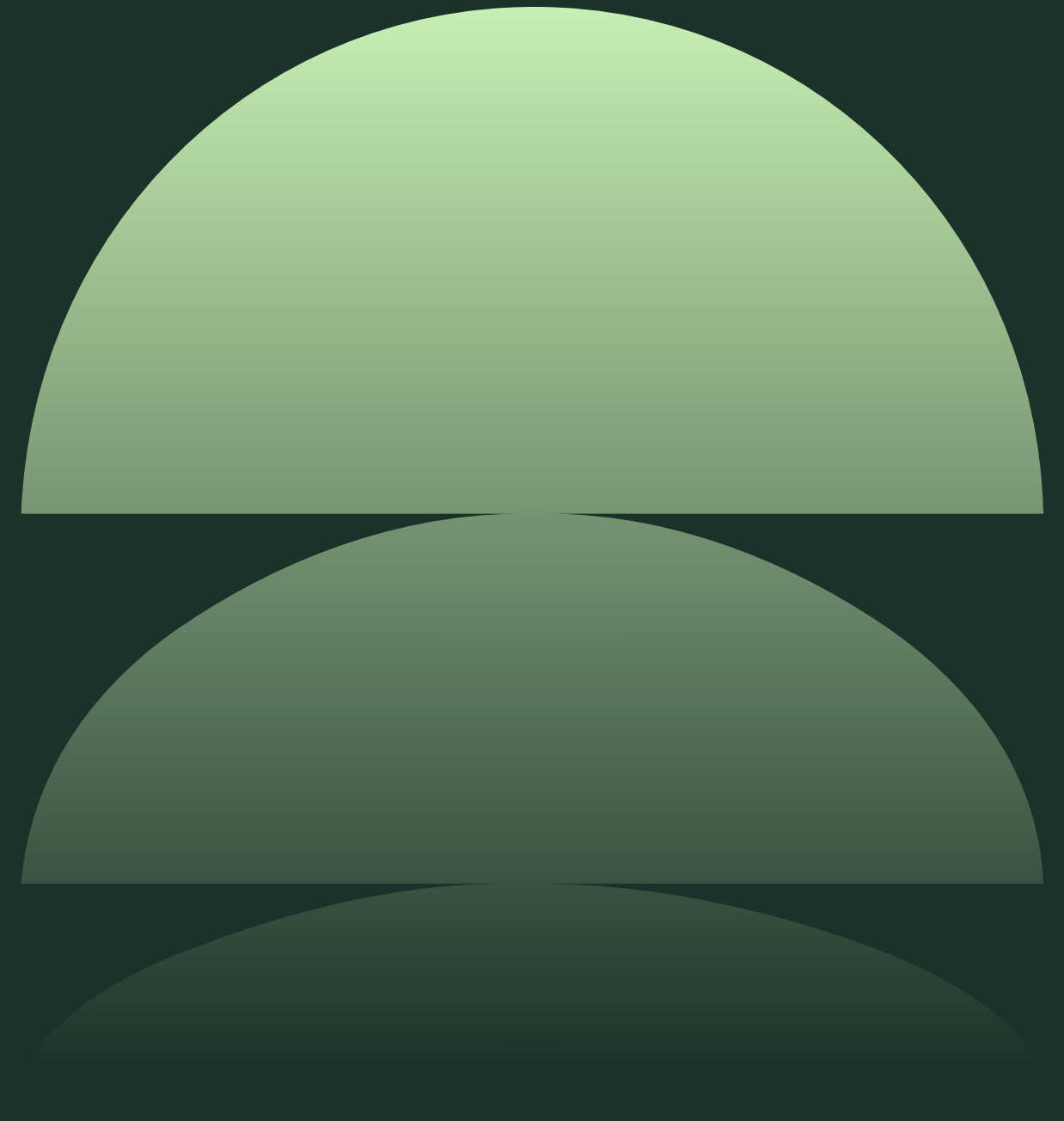


RETAIL REPORT 2027



TEAM 6

SEPTEMBER

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Agenda

01 **Introduction**

02 **Executive Summary**

03 **Problem Statement**

04 **Sales by Region**

05 **Top Product Performance**

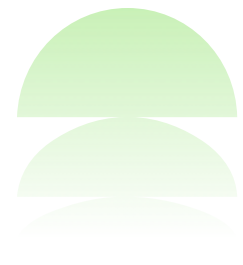
06 **Sales By Category**

07 **Customer Segments**

08 **Geography**

09 **Forecast**

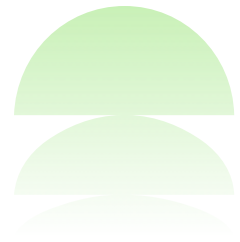
10 **Conclusion & Next Steps**



Introduction

Hello everyone, and a special welcome to our instructor, Eng. Hagar. Have you ever wondered how retail businesses generate millions in sales, yet still end up losing money? We asked ourselves this exact question. My name is Mohamed Khalid, and with my talented team of Ahmed, Mariem, and Youssef, we are excited to present our final project: the Sales Data Analysis and Profit Prediction App. In this project, we didn't just look at past sales; we built a complete pipeline to forecast profitability.



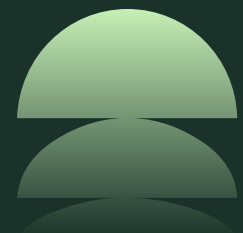


Executive Summary

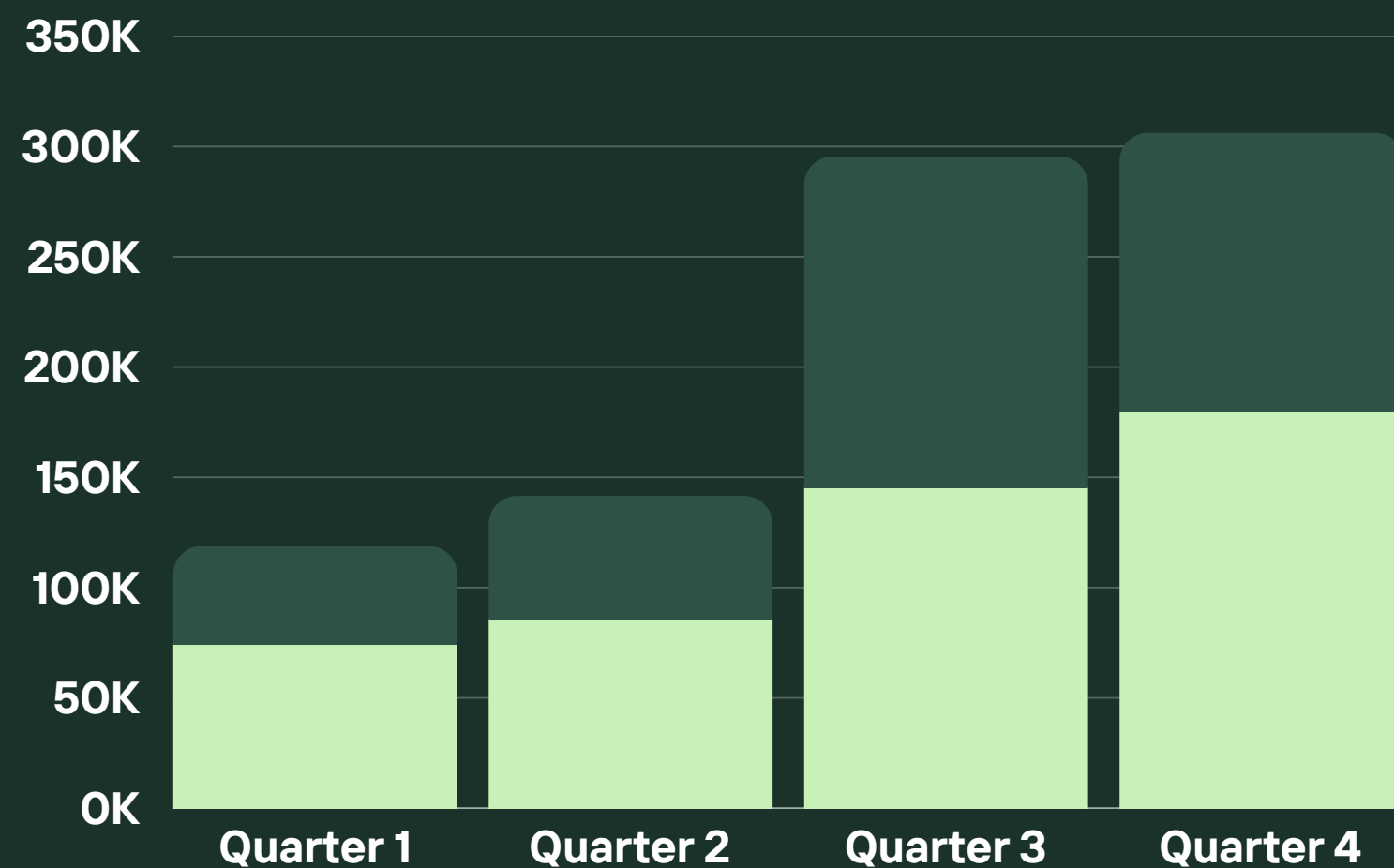
01

The Business Question: How can we accurately predict transaction profitability and identify the key drivers of profit and loss across different retail categories and regions?

Key Finding & Recommended Action: The analysis revealed that while Furniture generates high revenue, it yields extremely low profit margins due to high discount rates, whereas Technology (especially Copiers and Phones) drives the majority of net profit. A Random Forest model successfully predicted profit with 92.79% accuracy based on sales and cost data. Action: We recommend revising the discount strategies in underperforming states (like Texas and Ohio) and reallocating marketing efforts toward high-margin Technology products.



Quarterly Sales Performance



Revenue By Quarter

Q1

74.45 K

Steady start, supported by enterprise deals.

Q2

85.73 K

Moderate revenue growth supported by early seasonal campaigns.

Q3

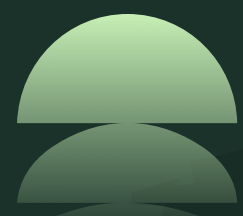
145.42 K

Significant sales surge accompanied by a sharp rise in promotional discounts.

Q4

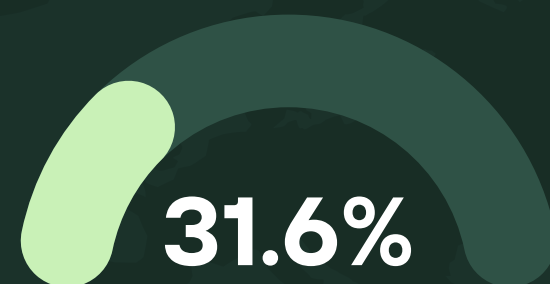
179.63 K

Peak revenue period driven by aggressive year-end holiday promotions and high discount volume.



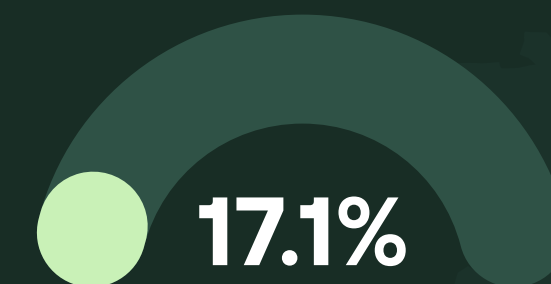
Sales by Region

Regional Sales Distribution



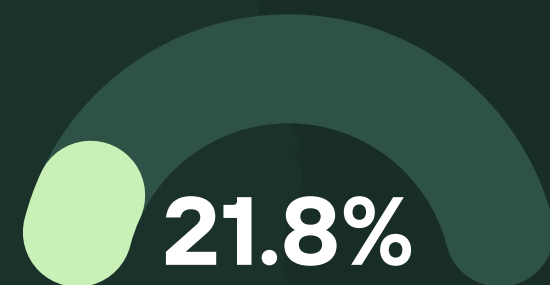
WEST

Continues to lead in revenue and customer base.



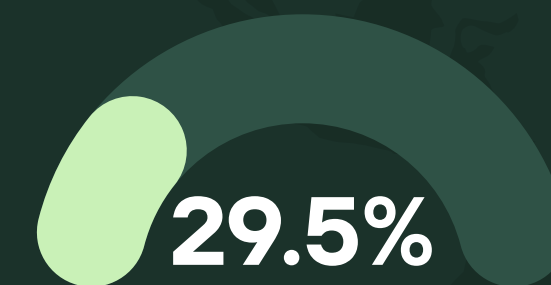
South

Stable growth with strong SMB engagement.



Central

Fastest-growing region with new partnerships.



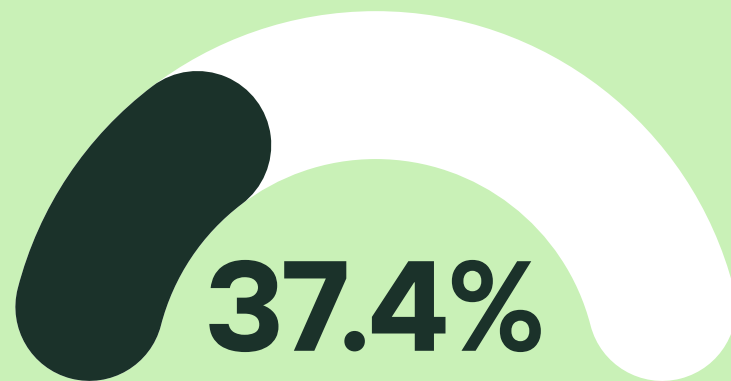
East

Merging markets with great expansion potential.



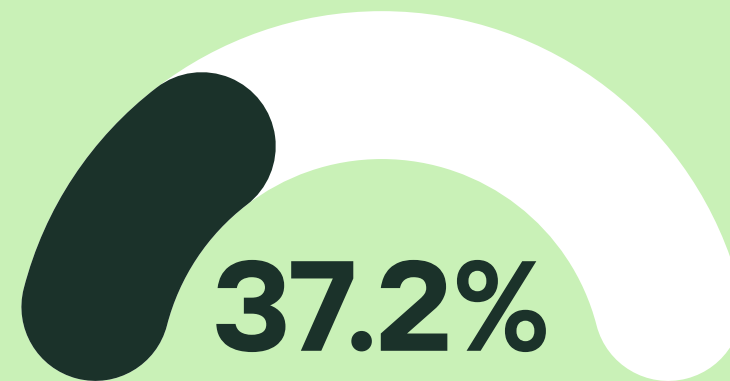
Top Product Performance

Product Revenue Distribution



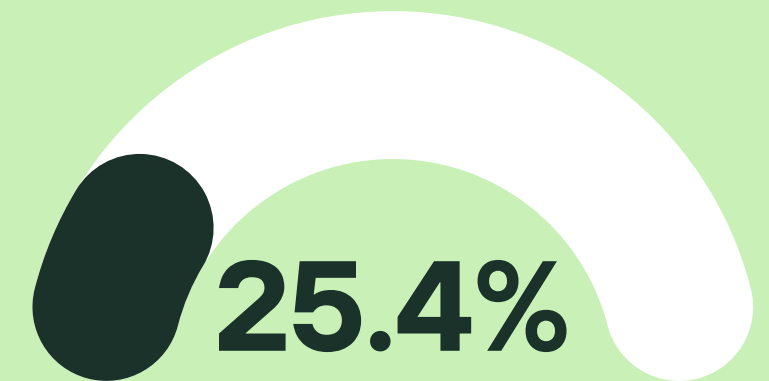
Phones

Best-selling item, especially in enterprise sector.



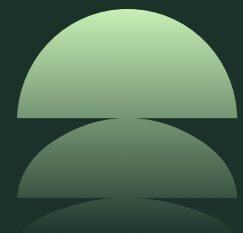
Chairs

Strong performer after recent updates.

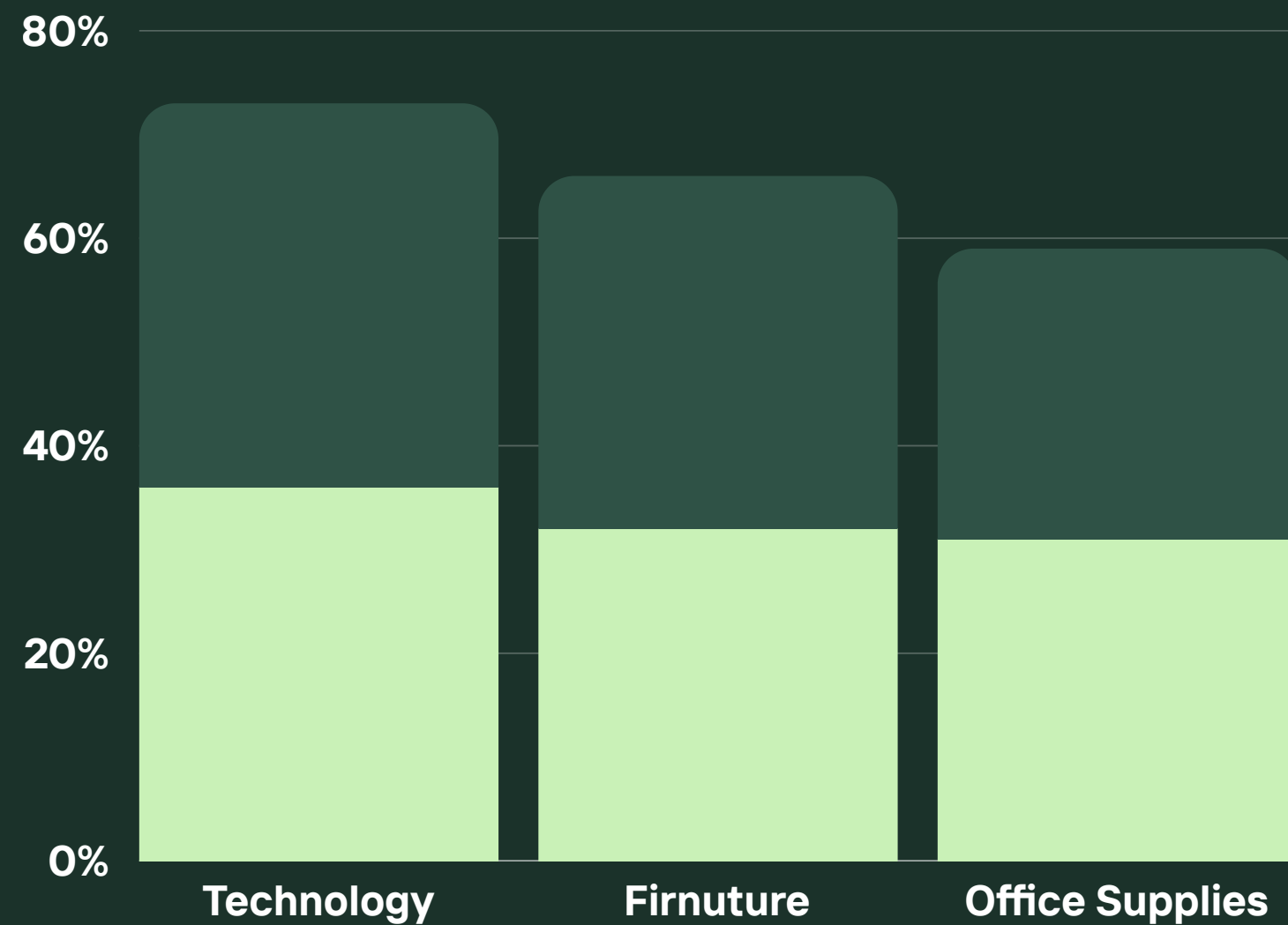


Storage

Rapid growth in mid-market segment.



Sales By Category



Categories Sales Comparison

36%

Technology

Impressive results achieved by Technology Category

32%

Furniture

Enhanced Sales growth.

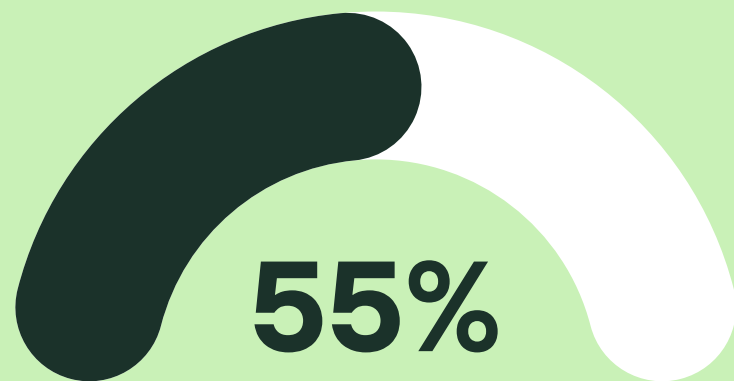
31%

Office Supplies

Good Growth

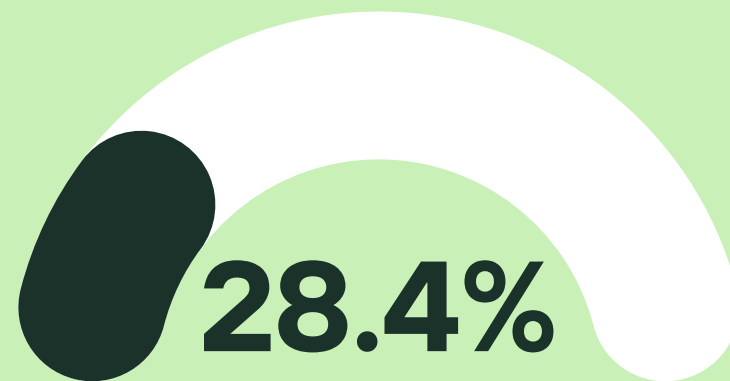


Customer Segments



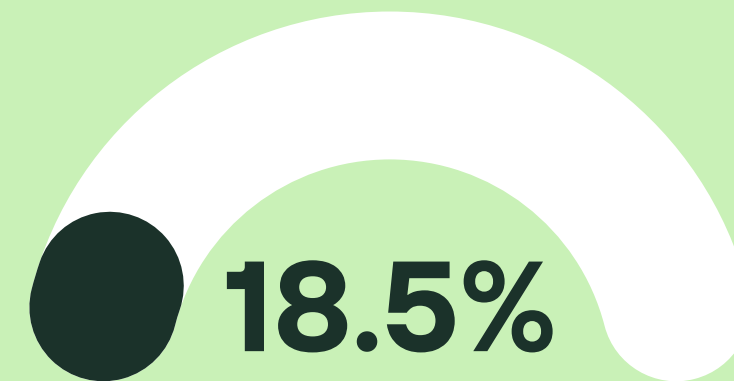
Consumer

Main revenue contributor with long-term contracts.



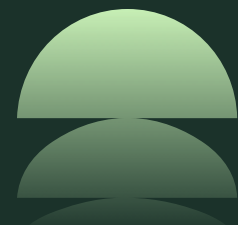
Corporate

Fast-growing group, boosted by digital campaigns.



Home Office

New market opportunity



Recommendations

01



Regional Discount Restructuring

Restricting excessive discounting in loss-making states will protect profit margins.

02



Furniture Cost Audit

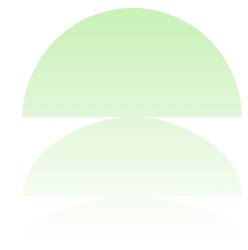
Reevaluating supply chain costs for low-margin items will improve category profitability.

03



Predictive Order Screening

Integrating the machine learning model at checkout flags unprofitable transactions early.



Forecast

01

**Data Preprocessing &
Pipeline Setup**

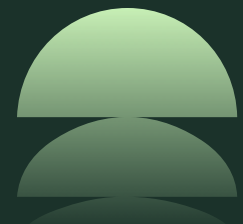
02

**Model Training &
Evaluation**

03

**Profit Prediction & App
Integration**





Tech Stack



01

Python : Cleaning, Exploring, Modeling

Pandas, Matplotlib, Seaborn, Plotly,
Stream

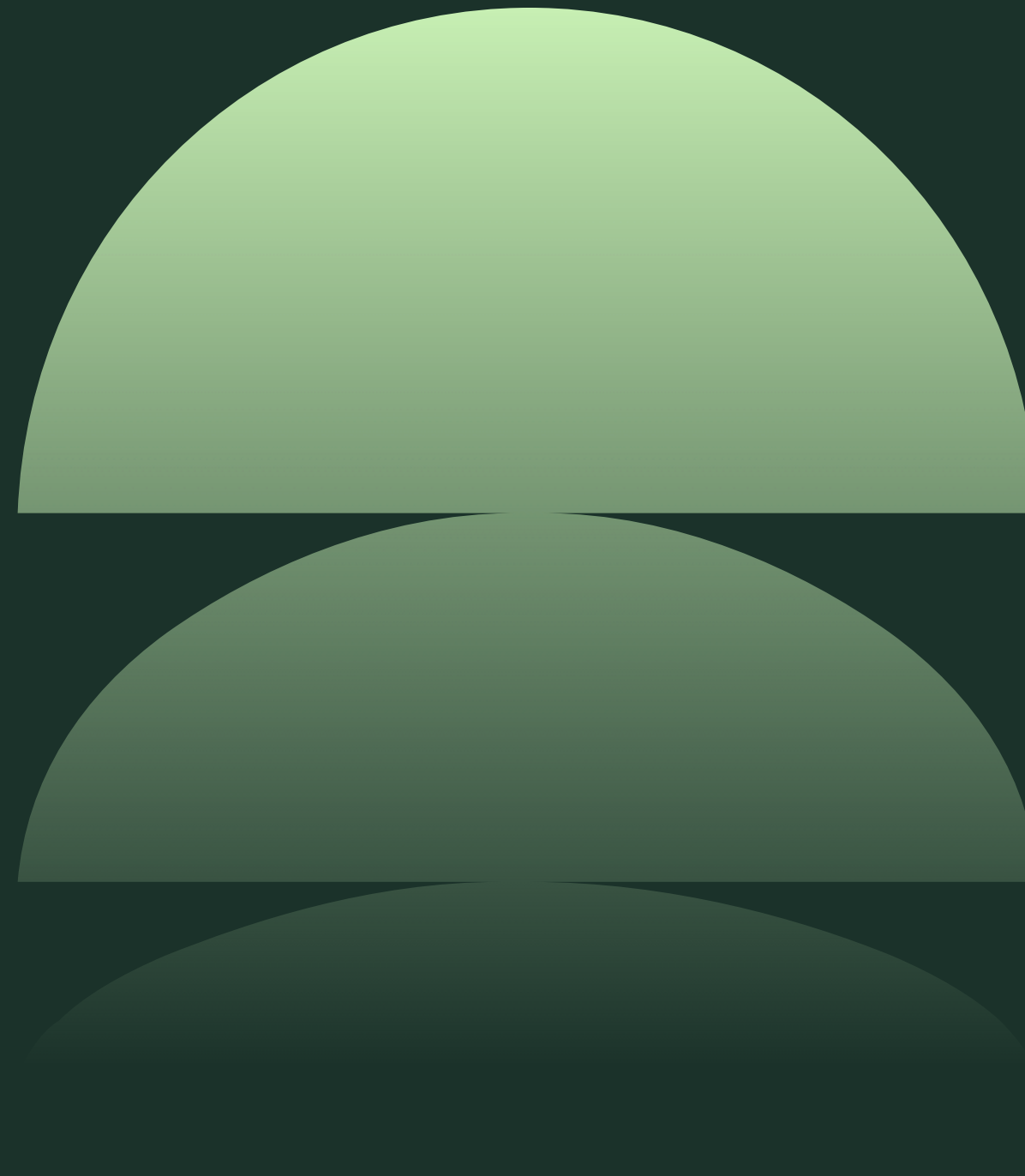
02

Power Bi : Visualization Tool

03

Deployment

THANK YOU



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